MIL-P-46212E (PA)
6 August 1975
SUPERSEDING
MIL-P-46212D (MU)
12 May 1969

MILITARY SPECIFICATION

PRIMER, ELECTRIC, M80A1
PARTS FOR

This specification is approved for use by all Department and Agencies of the Department of Defense.

1. Scope

1.1 This specification covers the parts for one type of electric primer designated as Primer, Electric, M80A1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550	- Ammunition and Special Weapons, General Specification for

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-1168	- Lot Numbering of Ammunition
MIL-STD-1169	- Packaging, packing and Marking for Shipment of Inert Ammunition Components

FSC: 1390

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MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ARMY

8831022 - Plug
8839500 - Body, Metal Parts Assembly

(Copies of specification, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other Publication.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING MATERIALS

ASTM-B-117-64-Salt Spray (FOG) Testing
(Copy may be purchased directly from the American Society for Testing Materials, 1916 Race St., Philadelphia, PA. 19103)

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Metal parts assembly and plug.-The metal parts assemblies and plugs shall comply with all requirements specified on Drawings (dwgs.) 8831022 and 8839500 and with all requirements specified in applicable specifications.

3.3 Hydrostatic test.-The primer tubing shall withstand the application of a minimum (min.) hydrostatic pressure without rupture or indication of flaws, as evidenced by leaks, and without permanent deformation beyond drawing requirements, when tested as specified in 4.4.1 (see 6.3).

3.4 Pull test of brazed joint.-The brazed joint securing the head to the primer body shall withstand a pull of 5500 pounds minimum without evidence of separation of the head from the primer body when tested as specified in 4.4.2.

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3.5 Workmanship.-All parts shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

3.6 First Article Inspection.-This specification makes provisions for first article inspection. Requirements for submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. In addition, the Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A2550 shall apply.

4.2 First Article Inspection

4.2.1 Submission.-The contractor shall submit a first article sample designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of 10 each individual components, which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirement of the contract and submit a record of this inspection with the sample, including statements of findings for materials, processes, and tests. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. A first

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article sample or portion thereof as directed by the contracting officer shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be performed.-Assemblies and components will be subjected by the Government to any or all of the examinations or tests specified in this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection.-If any assembly or component fails to comply with any or all of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of assembly component in the sample to comply with any or all of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article sample or portion thereof. Until a first article sample is accepted, the contractor is, in no way, authorized by the Government to initiate regular production unless otherwise directed by the contracting officer.

4.3 Inspection Provisions

4.3.1 Inspection Lot Formation.-The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions, and complies with the provisions for submission of product as specified in MIL-STD-105. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical

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defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Standard MIL-STD-1235 may be used if approved by the procuring activity. Sampling plans shall be applied to the individual characteristics listed using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Head (see dwg. 8839501 covering a detail of dwg. 8839500)

Categories	Defects	Method of Inspection
Critical: None defined		
Major: None defined		
Minor:		
	201. Total length	Gage
	202. Depth of body counterbore	Gage
	203. Evidence of poor workmanship (see 3.5)	Visual

4.3.2.2 Body (see dwg. 8839496 covering a detail of dwg. 8839500)

Categories	Defects	Method of Inspection
Critical:		
	1. One or more flash holes missing	Gage
Major:		
	101. Plug gage fails to pass through body	Gage
	102. Location of first row of holes	Gage
	103. Location of second row of holes	Gage
	104. Wall thickness	Gage
	105. Diameter of flash holes	Gage
	106. Pitch diameter of thread, maximum (max.)	Gage
	107. Minor diameter of thread, max.	Gage

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Minor: AQL 2.5 percent

201.	Total length	Gage
202.	Length of thread	Gage
203.	Minor diameter of thread, minimum (min.)	Gage
204.	Pitch diameter of thread, min	Gage
205.	Outside diameter	Gage
206.	Evidence of poor workmanship (see 3.5)	Visual

4.3.2.3 Plug (see dwg. 8831022)

Categories	Defects	Method of Inspection

Critical: None defined

Major:

101.	Pitch diameter of thread, min.	Gage
102.	Major diameter of thread, min.	Gage

Minor: AQL 2.5 percent

201.	Pitch diameter of thread, max.	Gage
202.	Major diameter of thread, max.	Gage
203.	Thickness	Gage
204.	Width of wrench slot, min.	Gage
205.	Depth of wrench slot	Gage
206.	Protective finish missing or damaged	Visual
207.	Evidence of poor workmanship (see 3.5)	Visual

4.3.2.4 Body, metal parts assembly (see dwg. 8839500 and dwg. 8839501 covering a detail of dwg. 8839500)

Categories	Defects	Method of Inspection

Critical:

1.	Crack or split (prior to plating)	Visual

Major:

101.	Pitch diameter of external head thread, min.	Gage
102.	Major diameter of external head thread, min.	Gage
103.	Pitch diameter of internal head thread, max.	Gage

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104. Minor diameter of internal head thread, max. .....Gage
105. Depth of internal head thread counterbore .....Gage
106. Braze not full 360 degrees .....Visual
107. Protective coating missing or damaged .....Visual
108. Concentricity of internal head thread with external head thread .Gage

Minor:

201. Length of external head thread, including thread undercut, min. ..Gage
202. Width of external head thread undercut, max. .....Gage
203. Depth of internal head thread, min. .....Gage
204. Diameter of external head thread undercut .....Gage
205. Diameter of internal head thread counterbore .....Gage
206. Width of wrench slots, min. (when applicable) .....Gage
207. Diameter of wrench holes, min (when applicable) .....Gage
208. Location of wrench holes (when applicable).....Gage
209. Depth of wrench holes (when applicable) .....Gage
210. Depth of wrench slots (when applicable) .....Gage
211. Inside diameter of head .....Gage
212. Outside diameter of head .....Gage
213. Thickness of flange .....Gage
214. Diameter of flange, max. .....Gage
215. Pitch diameter of external head thread, max. .....Gage
216. Major diameter of external head thread, max. .....Gage
217. Total length, max. .....Gage
218. Concentricity of internal head thread counterbore with internal head thread .....Gage
219. Concentricity of flange with external head thread .....Gage
220. Concentricity of head outside diameter with external head thread .....Gage

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221. Pitch diameter of internal head thread, min. .....Gage
222. Solder on interior surfaces .....Visual
223. Marking, missing, incorrect or unidentifiable .....Visual
224. Evidence of poor workmanship (see 3.5) .....Visual

4.3.3 Testing

4.3.3.1 Hydrostatic test (see 3.3) - Critical defect - This test shall be conducted 100 percent in accordance with the procedure specified in 4.4.1. If any primer tubing fails to comply with the requirement specified, it shall be classed defective and no primer bodies shall be fabricated therefrom.

4.3.3.2 Pull test of brazed joint (see 3.4) -Critical defect.

4.3.3.2.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified, 150 body assemblies shall be randomly selected from each lot for this test. If any body fails to comply with the applicable requirement the lot shall be rejected. The test shall be as specified in 4.4.2 using equipment in accordance with 4.3.4. The sample assemblies subjected to this test shall not return to the lot.

4.3.3.2.2 After three consecutive lots have met the criteria specified in 4.3.3.2.1, 50 body assemblies shall be randomly selected from each lot. If any body fails to comply with the applicable requirement, the lot shall be rejected. The test shall be performed as specified in 4.4.2 using equipment in accordance with 4.3.4. The Sample assemblies subjected to this test shall not return to the lot.

4.3.3.3 Salt Spray (Fog) Testing.-Major defect - A sample of five (5) primer bodies from each lot be submitted to a Salt Spray (Fog) Testing. The test shall be performed as specified in 4.4.3 Rejection criteria shall be as specified in the applicable finish specifications.

4.3.4 Inspection Equipment.-The inspection equipment required to perform the examinations and tests prescribed in this section is identified under the "Method of Inspection: headings in 4.3.2 and 4.3.3 (see 6.2).

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4.4 Test methods and procedures

4.4.1 Hydrostatic test.-Prior to machining, the primer tubing shall be tested for not less than five seconds at the minimum hydrostatic pressure as calculated by the following formula:

P=1.8 St divided by D

where:
P = tests pressure in pounds per square inch (psi)
S = minimum specified yield strength, psi
T = specified minimum wall thickness in inches
D = specified maximum outside diameter in inches

4.4.2 Pull test of brazed joint.-The body assembly shall be assembled to the applicable test fixture specified in 4.3.4. The specified load shall be applied in an axial direction. The rate of travel of the testing machine shall be 0.100 to 0.125 inch per minute. The sample assemblies subjected to this test shall not be returned to the lot.

4.4.3 Salt Spray. The primer bodies shall be tested in accordance with ASTM Standard Method of Salt Spray (Fog) Testing, ASTM Designation: B117-64. The primer bodies so tested shall be scrapped.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.-The parts shall be packaged, packed and marked in accordance with Standard MIL-STD-1169.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

c. First article inspection provisions.

6.2 Inspection Equipment Designs.-Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection"

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heading Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified dontractor designs of SIE in accordance with 6.2.2 and 6.2.3 should he elect to do so.

6.2.1 SIE Designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detailed design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.2.2 Contractor Designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor propses to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings of other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.2.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204 "Inspection equipment Design" for guidance. The item detail specification number, paragraph number, and defect number from Section 4 shall be referenced on each contractor design together with the component or assembly drawing number, revision letter and date to which the specific design applies.

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6.2.3 Submission of Designs for Approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements Lists (DD Form 1423). Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. Designs shall be submitted for approval to Commander, Picatinny Arsenal, ATTN: SARPA-QA-T-T, Dover, New Jersey 07801.

6.3 Hydrostatic Test.-If the primer tubing procured for the bodies manufactured in accordance with this specification has been previously subjected to the hydrostatic test specified in this specification, a second hydrostatic test will not be required.

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1390 - A096

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